10 SQL FUNCTIONS EVERY DATA ENGINEER USES DAILY

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1.COUNT()

Count Records Efficiently
Used to count rows, events, users, or transactions.

Use Cases:

- Data validation
- Record counting
- Pipeline monitoring

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

2.SUM()

Calculate Totals
Aggregates numerical values.

Use Cases:

- Revenue Reporting
- KPI Calculations
- Financial Metrics

```
SELECT SUM(order_amount) AS revenue
FROM orders;
```

3.AVG()

Calculate Average Values
Useful for trend and performance analysis.

Use Cases:

- Customer Analytics- Average of Customer Purchase
- Pricing Analysis- Avgerage of Product
- Performance Metrics - Average Performance of Dept

```
SELECT AVG(order_amount) AS
avg_order_value
FROM orders;
```

4.COALESCE()

Replace NULL Values
Returns the first non-null value.

Common Use Cases:

- Data cleaning
- Handling missing values
- ETL transformations

```
SELECT
 customer_name,
 COALESCE(phone,'Not Available') AS
phone
FROM customers;
```

5.CONCAT()

Combine Multiple Columns
Creates readable strings from multiple fields.

Use Cases:

- Full names
- Address formatting
- Reporting

```
SELECT
    CONCAT(first_name,' ',last_name)
AS full_name
FROM employees;
```

6.UPPER() / LOWER()

Standardize Text Data
Converts text to uppercase or lowercase.

Use Cases:

- Data standardization
- Case-insensitive matching
- Data quality improvement

```
SELECT
 UPPER(city) AS city_upper,
 LOWER(email) AS email_lower
FROM customers;
```

7. DATE_TRUNC()

Aggregate by Time Periods
Groups timestamps into day, month, quarter, etc.

Use Cases:

- Monthly reports
- Time-series analysis
- Dashboard metrics

```
SELECT
 DATE_TRUNC('month', order_date) AS
month,
 COUNT(*) AS orders
FROM orders
GROUP BY 1;
```

8.CAST()

Convert Data Types
Transforms one datatype into another.

Use Cases:

- Schema transformations
- Data warehouse loading
- ETL processing

```
SELECT
    CAST(order_amount AS
DECIMAL(10,2))
FROM orders;
```

9.SUBSTRING()

Extract Specific Parts of Text
Useful for parsing IDs and strings.

Use Cases:

- Data parsing
- Code extraction
- Log analysis

```
SELECT
 SUBSTRING(customer_id,1,3) AS
region_code
FROM customers;
```

10.CASE WHEN

Add Business Logic in SQL
Conditional transformation directly inside queries.

Use Cases:

- Customer segmentation
- Business rules
- Feature engineering

```
SELECT
 customer_name,
 CASE
 WHEN spend > 10000 THEN 'Gold'
 WHEN spend > 5000 THEN 'Silver'
 ELSE 'Bronze'
 END AS customer_tier
FROM customers;
```

MASTER THESE 10 FUNCTIONS FIRST. EVERY DATA ENGINEER USES THESE DAILY FOR:

DATA CLEANING
ETL PIPELINE
REPORTING
DATA WAREHOUSING
ANALYTICS ENGINEERING
DASHBOARD DEVELOPMENT

THANK YOU